Pattern-learning-based Noise Elimination Algorithm in Photoacoustic Sensing and Imaging

Zhi Tu, Boshi Wang, Tingyang Duan, *Student Member, IEEE,* and Fei Gao*, *Member, IEEE*

***Abstract*—As one of the fastest-growing imaging modalities in recent years, photoacoustic (PA) imaging has attracted tremendous research interest for various applications including anatomical, functional and molecular imaging. However, the PA signal's amplitude is usually quite weak and can be easily distorted by instrumental noise and interference, which can severely degrade the image quality. To improve the PA signal's signal-to-noise ratio efficiently, this paper introduces a pattern-learning based PA (PLPA) detection method to eliminate the periodically interference noise for PA sensing and imaging. Both simulation and experimental results are demonstrated to prove the validity of the proposed algorithm.**

I. Introduction

Photoacoustic (PA) imaging as a noninvasive imaging technique based on photoacoustic effect [1-9] attracted tremendous research interest for applications [10-11] including anatomical, functional and molecular imaging. The PA signal's amplitude is usually quite weak, and many applications of PA imaging technique have quite high demand for the signal-to-noise ratio (SNR) of PA signals, while unfortunately the instrumental noise and interference can greatly degrades the PA signal quality. To eliminate the interference of noise in PA sensing and imaging, there are many research work done regarding algorithms improvement from variant aspects including the sensor design, the signal processing, and the image processing [12-14]. Regarding signal processing, there are also various means to reduce noises such as by improving filter design [15-20] to reduce high frequency noises.

Unlike white noise, the noises introduced by equipment's interferences, such as laser firing, are periodic signals independent with the PA signal but have random phase. The frequency spectrum of these interference noise is likely to be overlapped with PA signals' spectrum, so applying low-pass or band-pass filter is not an effective option. Since the random phase of the interference noise conforms to Uniform Distribution, averaging repeated samples is an effective way and can eliminate the white noises at the same time. However, data averaging requires a huge amount of sample measurements for every single point, which is too time-consuming to be practical. To overcome the above-mentioned problem, in paper, we introduce a pattern-learning based PA (PLPA) detection method to eliminate the periodically interference noise in PA sensing and imaging. This algorithm can automatically learn the pattern of the equipment's interference noise and apply the pre-learned pattern to eliminate the interference of equipment noise with low time complexity and good accuracy.

Zhi Tu, Boshi Wang, Tingyang Duan and Fei Gao are with the School of Information Science and Technology, ShanghaiTech University, Shanghai 201210, China (*corresponding author: gaofei@shanghaitech.edu.cn).

II. Algorithm

The principle of the proposed PLPA algorithm is shown in Fig. 1. which basically consists of two major steps: signal pattern learning and noise elimination.

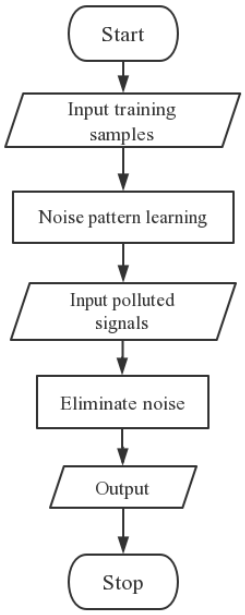

Figure 1. Flow chart of the proposed PLPA algorithm.

The interference noise has a stable pattern with a certain periodicity, but a random phase information, we can assume that for a raw noisy PA signal, the phase of the interference noise signal follows uniform distribution overlapping the true PA signal. Then we could assume that the raw noisy PA signal consists of the pure PA signal, the periodic interference noise and white noise, which can be expressed as:

$$f(t) = g(t) + h(t+\epsilon) + w(t), \epsilon \sim Unif(0, L) \quad (1)$$

where $f(t)$ is the measured raw noisy PA signal, $g(t)$ is the original pure PA signal, $h(t+\epsilon)$ is the interference noise whose phase term ϵ is random and conforms to Uniform distribution, and $w(t)$ is white noise.

To learn the pattern of interference noise $h(t+\epsilon)$, the proposed PLPA algorithm needs to firstly extracts the pattern of the interference noise signal $h(t)$ from the raw PA signal

$f(x)$. Considering the randomness of both interference and white noise, the averaging procedure is needed to eliminate the influence of the noise and lead to a constant.

$$\lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} h(t+\epsilon_i) = C \tag{2}$$

$$\lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} w_i(t) = 0 \tag{3}$$

Therefore, if we keep the $g(t)$ same, which means the training PA signals are collected from the same point, and the scale of training samples is large enough, then we have:

$$\lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} f_i(t) = F(t) = g(t) + C \tag{4}$$

We subtract a constant C to make $F(0) - C = 0$, then we obtain the pure signal $g(t)$. This is also the method used in many conventional white noise elimination methods, which is valid but too time-consuming due to multiple data averaging. Then for every $f(t)$ in the training samples, we can subtract the pure signal $g(t)$ to get $h(t+\epsilon_i) + w_i(t)$, which is shown in Eq. (5):

$$f_i(t) - g(t) = h(t+\epsilon_i) + w_i(t) \tag{5}$$

For every $h(t+\epsilon_i)$, it is easy to find out its phase position. The we could adjust all the $h_i(t)$ to the same phase position and average to eliminate the influence of white noises to obtain $h(t)$, which is the periodic noise, as shown in Eq. (6).

$$\lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} h(t) + w_i(t) = h(t) \tag{6}$$

Then when a raw noisy PA signal is collected, the PLPA algorithm will locate the phase position of the interference noise first and subtract it by the pre-learned noise pattern.

According to Eq. (3), it deserves noting that this problem can be solved by barely averaging, but that would acquire a large amount of data sampling at every scanning point that is too time consuming. On the other hand, our proposed PLPA algorithm only needs one averaging procedure in the pattern pre-learning stage, which largely reduce the time consumption.

III. Simulation Result

In simulation, we generate raw noisy signal including random phase position to simulate the periodic interference noise $h(t+\epsilon)$, one exponential function to simulate the pure signal $g(t)$ and a random function to simulate the white noise $w(t)$. In the simulation, the analytical equations are shown in Eq. (7)-(10):

$$h(t) = \sin\big(2(t-\epsilon)\big), \epsilon \sim Unif(0,\pi), t\,\epsilon[0,10] \tag{7}$$

$$g(t) = 2e^{|t-1|}, t\,\epsilon[0,10] \tag{8}$$

$$w(t) = random(0,0.4) \tag{9}$$

$$f(t) = h(t) + g(t) + w(t), t\,\epsilon[0,10] \tag{10}$$

In the simulation, the scale of training set is 200 samples. The simulation results are shown in Fig. 2. Fig. 2(a) shows the average of all the training set $f(t)$. The curve still has a little errors because of the limit of the scale that cannot completely eliminate the influence of white noises. Fig. 2(b) shows the average of the interference noises. Fig. 2(c) plots a test signal and Fig. 2(d) is its corresponding output signal . From Fig. 2(e), we can see that the difference between the enveloping curve of output signal and the pure signal is small enough. Though in the process of eliminating the noise, the white noise remains, the signal used for imaging would be the enveloping curve of the output signal, which would eliminate the influence caused by the white noise as Fig. 2(e) shows. Fig. 2(f) is its enveloping curve.

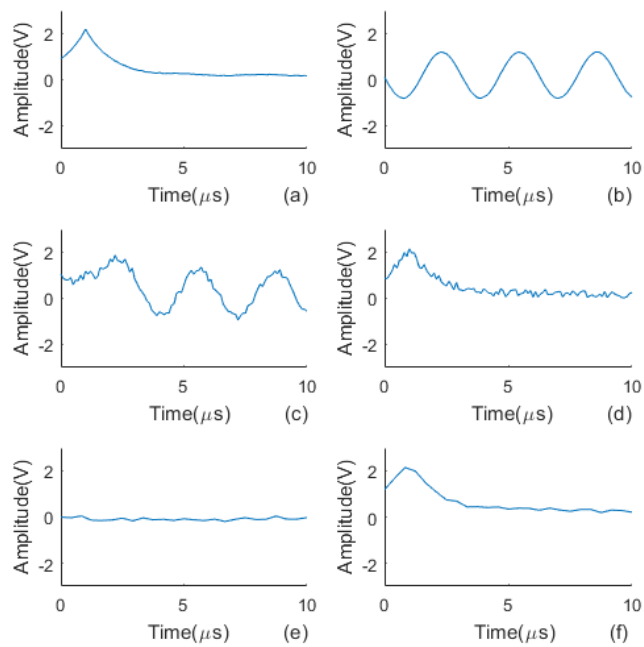

Figure 2. (a) The average signal of the total 200 samples. (b) The derived noise signal wave. (c) A test simulate signal. (d) The test result corresponding to (c). (e) The difference between the enveloping curve of output signal and the generated target signal. (f) The envelope curve of the output signal.

IV. Experimental Setup and Results

A. Setup

The experimental setup to prove the effectiveness of the algorithm is similar with the simulation setup as Fig. 3 shows. The generated laser travelled through fiber and condenser lens and then was focused on a 3cm long, black ink filled tube with diameter 4mm at the bottom of a 5cm deep water tank. Then the PA signals were generated from the tube and collected by an ultrasound transducer. After amplification by a low-noise amplifier, the PA signal was digitized by an oscilloscope and then collected in computer. To emulate the tough situation, the experiment also only collected 200 training samples.

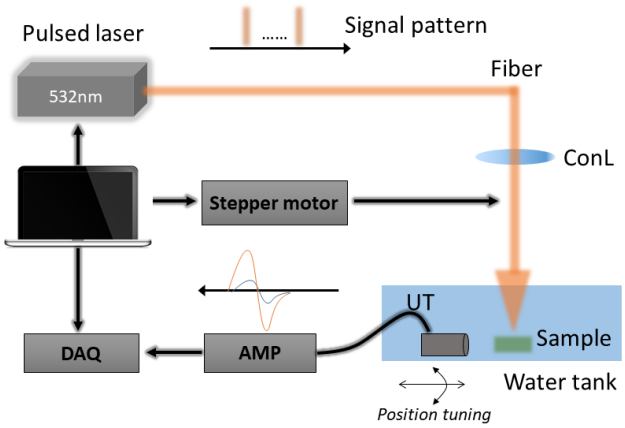

Figure 3. The schematic of the experimental setup. ConL: condenser lens, UT: ultrasound transducer, DAQ: data acquisition, AMP: amplifier

B. *Experimental Results*

The experimental results are shown in Fig. 4. We evaluate the first 4 us for each PA signal that is normally considered coming from the object.

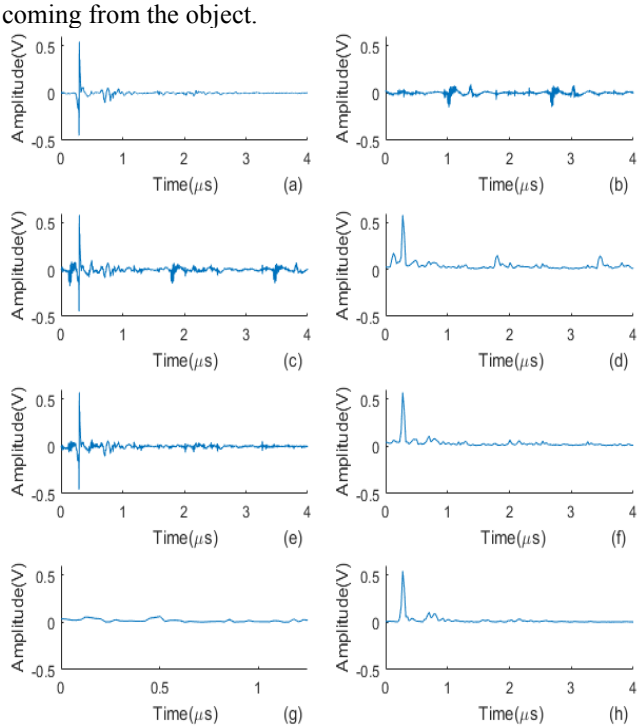

Figure 4. (a) The derived pure signal. (b) The calculated noise signal. (c) A test sample input. (d) The enveloping curve of the test sample. (e) The output signal corresponding to (c). (f) The enveloping curve of the output signal. (g) The comparative figure, which is the absolute value of subtracting (f) from (b). (h) The enveloping curve of the pure signal.

Fig. 4(a) shows the derived pure signal curve by averaging the training samples, and Fig. 4(b) is the extracted interference noise pattern. Fig. 4(c) is a measured noisy PA signal and Fig. 4(d) is its enveloping curve. Fig. 4(e) is the output signal and Fig. 4(f) is its enveloping curve. Fig. 4(h) is the enveloping curve of Fig .4(a). As shown in Fig. 4(g), the comparative figure between the enveloping curve of the original signal and the output signal shows that the amplitude difference between those in time domain is a curve lying around zero. Quantitatively, as Table. 1 shows, the output signal has nearer value to the pure signal in both maximum value and mean. The curve subtracting output from pure signal has small enough mean and variance, which means that the difference between the two signals is small enough and the result is stable, indicating the excellent noise suppression performance of the proposed PLPA algorithm.

Table. 1. The comparative table about maximum value, mean, and variance for the signals in the first $1\mu s$.

	Maximum	Mean	Variance
Pure signal enveloping curve	0.5403V	0.04646V	0.0081
Noised signal enveloping curve	0.5811V	0.0705V	0.0088
Output signal enveloping curve	0.5673V	0.0615V	0.0081
The subtractive curve in Fig. 4(g)	0.0617V	0.0175V	2.4964×10^{-4}

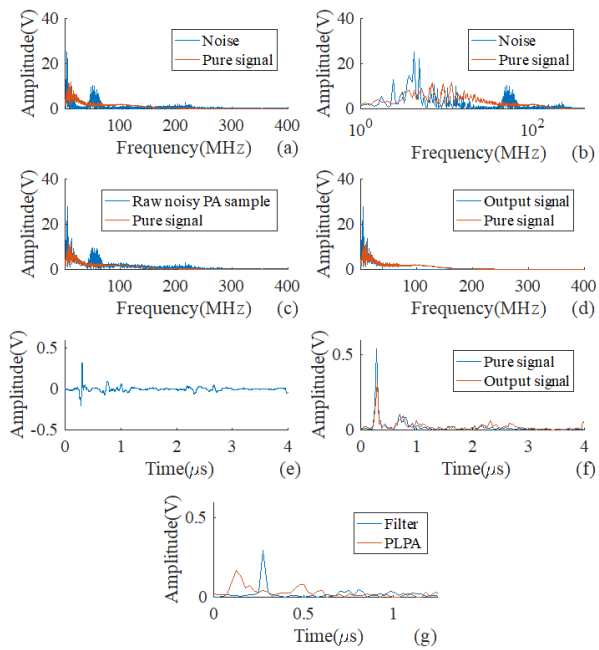

Figure 5. (a) The frequency domain figure of the derived noise signal and the pure signal. (b) The frequency domain semilogx figure of the derived noise signal and the pure signal. (c) The frequency domain figure of the raw noisy PA signal and the derived pure signal. (d) The frequency domain figure of the output signal and pure signal. (e) The time-domain figure of the output signal. (f) The enveloping curve of the derived pure PA signal and the output signal. (g) The absolute value of subtracting the enveloping curve of the output of the two methods from the enveloping curve of pure PA signal (The first $1.25\mu s$).

The frequency domain analysis is shown in Fig. 5. As Fig. 5(b) shows, in the low frequency part, the noise signal has two congregations and one of them overlaps the pure signal, which means if using filter, half of the congregation of the noise signal cannot be eliminated. The results of post-filter signal are shown in Fig. 5(c)-(f). From Fig. 5(e)-(f) we can notice that the interference of the patterned equipment generated noise remains (from $2\mu s$ to $3\mu s$) and the contrast ratio has obviously declined, and the output signal actually has several distortions. Fig. 5(g) shows that the result of

PLPA is closer to the pure signal than filtering. Therefore, we can conclude from Fig. 5 that the PLPA is more stable and efficient than using filter to eliminate periodic interference noises in photoacoustic imaging.

V. Discussion and Conclusion

Regarding the applications of the proposed PLPA algorithm for photoacoustic imaging, it could be expected that this algorithm will be feasible to remove interference noise with stable patterns, such as the interference from laser firing. Moreover, some other applications like noise without periodicity but with strong pattern are also feasible by some modification of learning methods in this algorithm. Last but not least, the recently developed nonlinear PA imaging may also benefit from the proposed PLPA algorithm [21-26], which usually requires higher detection sensitivity lower noise floor than conventional linear PA imaging.

In conclusion, the Pattern Learning based noise elimination Algorithm has been modelled, simulated and experimentally validated. The advantage of this algorithm has shown a much less time consumption than the conventional averaging methods and a noticeable better effect than filters. The merit of this algorithm may enable more applications for photoacoustic imaging with poor signal SNR. Further application of the noise elimination algorithm with higher accuracy and reliability and lower time complication and lower learning cost would promote the application of photoacoustic imaging in the future.

References

[1] Gao, F., Feng, X. & Zheng, Y. Advanced photoacoustic and thermoacoustic sensing and imaging beyond pulsed absorption contrast. Journal of Optics 18, 074006 (2016).

[2] Beard, P. Biomedical photoacoustic imaging. Interface Focus 1, 602-631, doi:10.1098/rsfs.2011.0028 (2011).

[3] Wang, L. V. Multiscale photoacoustic microscopy and computed tomography. Nature Photonics 3, 503-509, doi:10.1038/nphoton.2009.157 (2009).

[4] Xu, M. H. & Wang, L. H. V. Photoacoustic imaging in biomedicine. Review Of Scientific Instruments 77, 041101, doi:10.1063/1.2195024 (2006).

[5] Ntziachristos, V., Ripoll, J., Wang, L. H. V. & Weissleder, R. Looking and listening to light: the evolution of whole-body photonic imaging. Nature Biotechnology 23, 313-320, doi:10.1038/nbt1074 (2005).

[6] Yao, J. J. et al. High-speed label-free functional photoacoustic microscopy of mouse brain in action. Nat Methods 12, 407-+, doi:10.1038/nmeth.3336 (2015).

[7] Zhang, H. F., Maslov, K., Stoica, G. & Wang, L. V. Functional photoacoustic microscopy for high-resolution and noninvasive in vivo imaging. Nature Biotechnology 24, 848-851, doi:10.1038/nbt1220 (2006).

[8] Fei Gao, Xiaohua Feng and Yuanjin Zheng, "Micro-Doppler photoacoustic effect and sensing by ultrasound radar ", IEEE Journal of Selected Topics on Quantum Electronics, 22, 681806, 1-6 (2016)

[9] Fei Gao, Xiaohua Feng, Yuanjin Zheng, "Coherent photoacoustic-ultrasound correlation and imaging", IEEE Transactions on Biomedical Engineering, 61, 2507-2512 (2014)

[10] Fei Gao, Qiwen Peng, Xiaohua Feng, Bo Gao, and Yuanjin Zheng, “Single-wavelength blood oxygen saturation sensing combining optical scattering and absorption”, IEEE Sensors Journal, 16, 1943-1948 (2016)

[11] Xiaohua Feng, Fei Gao, Yuanjin Zheng, "Photoacoustics Based Close-loop Control for Nanoparticle Hyperthermia", IEEE Transactions on Biomedical Engineering, 62, 1728 - 1737 (2015)

[12] Dong, Yiqiu, and S. Xu. "Xu, S.: A New Directional Weighted Median Filter for Removal of Random-Valued Impulse Noise. IEEE Signal Processing Letters 14(3), 193-196." IEEE Signal Processing Letters 14.3(2007)

[13] Malfait, M, and D. Roose. "Wavelet-based image denoising using a Markov random field a priori model." IEEE Transactions on Image Processing A Publication of the IEEE Signal Processing Society 6.4(1997)

[14] Chan, Raymond H., C. W. Ho, and M. Nikolova. "Salt-pepper noise removal by median-type noise detectors and detail-preserving regularization. IEEE Trans Image Process." IEEE Transactions on Image Processing 14.10(2005):1479-1485.

[15] Shin, Hyunchol, and Y. Kim. "A CMOS Active-RC Low-Pass Filter With Simultaneously Tunable High- and Low-Cutoff Frequencies for IEEE 802.22 Applications." IEEE Transactions on Circuits & Systems II Express Briefs 57.2(2010):85-89.

[16] Li, Jing, et al. "A co-design study of low noise amplifier and band-pass filter." International Conference on Microwave and Millimeter Wave Technology IEEE, 2012:1-3.

[17] Yoon, Ki Cheol, et al. "Low noise and high gain dual-band active band-pass filter with GaAs MESFET using CRLH metamaterial." Microwave Conference Proceedings IEEE, 2014:697-699.

[18] Chun, Young Hoon, et al. "Design of an RF low-noise bandpass filter using active capacitance circuit." IEEE Transactions on Microwave Theory & Techniques 53.2(2005):687-695.

[19] Ahn, Dal, et al. "A design of the low-pass filter using the novel microstrip defected ground structure." IEEE Transactions on Microwave Theory & Techniques 49.1(2001):86-93.

[20] Fei Gao, Xiaohua Feng, Ruochong Zhang, Siyu Liu, Yuanjin Zheng, "Adaptive photoacoustic sensing using matched filter", IEEE Sensors Letters, 1, 3501003 (2017)

[21] Ruochong Zhang, Fei Gao, Xiaohua Feng, Siyu Liu, Ran Ding and Yuanjin Zheng, “Photoacoustic resonance imaging”, IEEE Journal of Selected Topics in Quantum Electronics, in press (2018)

[22] Fei Gao, Rahul Kishor, Xiaohua Feng, Siyu Liu, Ran Ding, Ruochong Zhang, Yuanjin Zheng, “An analytical study of photoacoustic and thermoacoustic generation efficiency towards contrast agent and film design optimization”, Photoacoustics, 7, 1-11 (2017)

[23] Fei Gao, Xiaohua Feng, Ruochong Zhang, Siyu Liu, Ran Ding, Rahul Kishor, Yuanjin Zheng, “Single laser pulse generates dual photoacoustic signals for differential contrast photoacoustic imaging”, Scientific Reports, 7, 626 (2017)

[24] Fei Gao, Ruochong Zhang, Xiaohua Feng, Siyu Liu, Ran Ding, Rahul Kishor, Lei Qiu and Yuanjin Zheng, “Phase-domain photoacoustic sensing”, Applied Physics Letters, 110, 033701 (2017)

[25] Fei Gao, Linyi Bai, Siyu Liu, Ruochong Zhang, Xiaohua Feng, Yuanjin Zheng, and Yanli Zhao, “Rationally Encapsulated Gold Nanorods Improving both Linear and Nonlinear Photoacoustic Imaging Contrast in vivo”, Nanoscale, 9, 79-86 (2017)

[26] Fei Gao, Yuanjin Zheng, Xiaohua Feng, and Claus-Dieter Ohl, “Thermoacoustic resonance effect and circuit modelling of biological tissue,” Applied Physics Letters 102, 063702 (2013)